

ITAS-MM: An Intelligent Multimodal Traffic Assistance System for Real Time Traffic Sign Recognition, Lane Detection, and Automatic Navigation

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Abstract—Traffic-related fatalities continue to claim roughly 1.19 million lives each year worldwide, as documented in the 2023 WHO report [1]. Motivated by this ongoing crisis, we design ITAS-MM — a driver assistance framework that unifies camera-based perception, spatial positioning via GPS, voice-driven interaction, and intelligent legal reasoning into a single coherent pipeline. Our work targets two complementary perception objectives: recognizing Vietnamese regulatory signage and maintaining lane awareness in real driving scenes. The processing chain combines YOLOv11 for sign classification and Ultra-Fast Lane Detection (UFLD) for lane geometry [2], [3], feeds the resulting events into a Retrieval-Augmented Generation (RAG) module backed by a Large Language Model (LLM) that retrieves and interprets the relevant clauses from Vietnamese traffic law (QCVN 41:2024 and Nghị định 168/2024/NĐ-CP) [4], [5], and produces spoken guidance through NVIDIA Parakeet CTC 0.6B [6]. On our Vietnamese sign corpus, the tuned YOLOv11 attains $mAP_{50} = 86.03\%$ with $Precision = 87.69\%$.

Index Terms—Traffic Sign Recognition, Ultra-Fast Lane Detection, RAG, LLM, Multimodal Feedback, YOLOv11, Fuzzy Inference System, Embedded AI, ADAS, TensorRT, Parakeet CTC

I. INTRODUCTION

A. Background and Motivation

Globally, traffic collisions remain a dominant cause of preventable death; WHO data ranks road injuries as the top killer of young people between 5 and 29 years old, with the burden falling disproportionately on developing nations [1]. Vietnam’s road network presents a uniquely demanding environment: motorcycles, cars, trucks, and pedestrians jostle for space on narrow two-lane streets, producing complex visual scenes that most commercial Advanced Driver Assistance Systems (ADAS) were never engineered to handle.

Although contemporary ADAS platforms have advanced considerably, three recurring shortcomings limit their effectiveness in practice:

- 1) **Single-channel alerting.** The majority of deployed systems rely solely on visual cues — dashboard icons, head-up display overlays, or screen pop-ups — forcing drivers to divert their gaze at precisely the moments when attention should remain on the road [7]. Politis *et al.* showed that combining auditory and haptic channels with visual warnings cuts reaction latency significantly.
- 2) **Absence of regulatory reasoning.** Existing detectors output bare category labels such as “Speed Limit 40” without connecting that label to the jurisdiction-specific rule it encodes. Under Vietnam’s QCVN 41:2024 standard [4], a single sign can carry different implications depending on whether the vehicle is in an urban, suburban, or highway zone — context that no current detector supplies.

B. Literature Review

Traffic Sign Detection and Recognition (TSDR). Deep-learning-based detectors have become the prevailing approach for real-time TSDR. Boukamcha [8] benchmarked YOLOv10 against YOLOv8, establishing throughput–accuracy trade-off baselines that subsequent work has built upon. Focusing specifically on Vietnamese road conditions, Nguyen *et al.* [9] undertook a controlled empirical study and reported that the YOLOv10-Nano variant reaches $mAP_{50} = 88.2\%$ on domestic sign classes while sidestepping the NMS latency penalty that burdens earlier architectures.

On the legal-understanding front, Pham *et al.* [10] introduced LexiSignVQA, a pipeline that answers regulatory questions about detected signs without task-specific fine-tuning,

and Huynh et al. [11] constructed a graph-structured retrieval engine for querying the Vietnamese traffic code through multimodal inputs.

Lane Detection. Qin et al. [2] reframed lane localization as a per-row selection task driven by structural priors, producing the Ultra-Fast Lane Detection (UFLD) framework whose throughput far exceeds dense-prediction alternatives. Their follow-up, UFLD v2 [3], introduced hybrid anchor-driven ordinal classification and achieved $F_1 = 75.0\%$ on CULane and 96.11% accuracy on TuSimple using a ResNet-18 backbone. Independently, Tsai et al. [12] validated that deterministic lane detection can run at real-time rates even on CPU-only embedded boards, reinforcing the viability of edge deployment.

Multimodal Driver Assistance. Multimodal AI for driving has grown rapidly. Xu et al. [13] demonstrated that large language models can generate context-aware spoken warnings. The NaviBlind project [14] proved that coupling PhoWhisper speech recognition with Gemini Flash enables effective voice-guided navigation. At HUST, Tri et al. [15] fused gaze-tracking data with steering-wheel telemetry to assess driver alertness, reporting 94.6% classification accuracy.

C. Limitations of Current Approaches

Two specific deficiencies motivate the present work:

Unreliable small-object detection. Signs whose projected area falls below roughly 20×20 pixels are routinely missed [9]. On crowded Vietnamese streets, where signs are often distant or partially hidden behind vehicles, this weakness translates directly into elevated false-negative rates.

No legal grounding. Raw class labels (e.g., “P.127a”) carry no regulatory meaning unless mapped to the applicable traffic law clause and qualified by zone context [4].

D. Contributions

To address these gaps, we make four contributions:

- **A parallel multi-task perception pipeline** that jointly handles sign detection and lane tracking while preserving the modular structure required for real-time driver assistance.
- **A RAG-powered legal reasoning engine** that retrieves and interprets Vietnamese traffic regulations (QCVN 41:2024, Nghị định 168/2024/NĐ-CP) through a Large Language Model, transforming raw detections into legally grounded advice.
- **A Vietnamese voice interaction layer** built on NVIDIA Parakeet CTC 0.6B [6] for speech recognition and a quantized TTS engine for spoken output, enabling hands-free driver interaction.
- **A multimodal interaction design** that couples visual, auditory, and geospatial channels, reducing the cognitive overhead on the driver.

The remainder of this paper is organized as follows. Section 2 details each pipeline stage. Section 3 reports quantitative experiments. Section 4 interprets the results and acknowledges

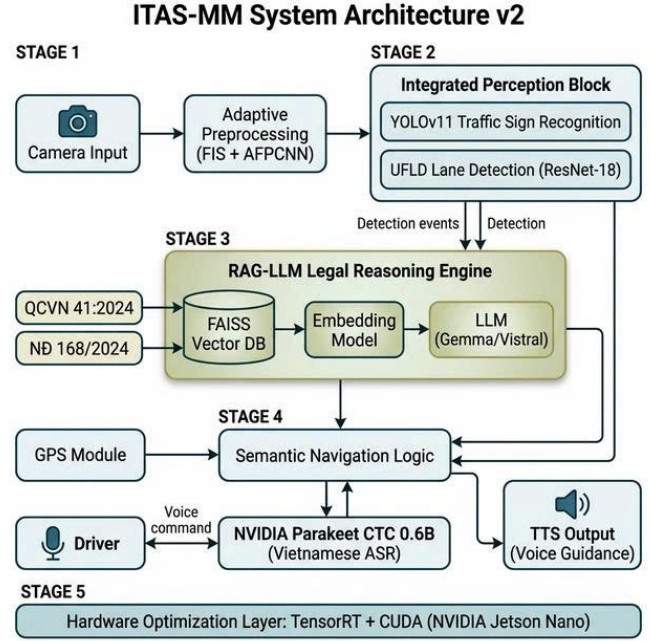


Fig. 1: The overall system architecture of ITAS-MM, showing the four-stage processing pipeline: Camera Input $\rightarrow$ Integrated Perception Block (YOLOv11 TSR + UFLD Lane Detection) $\rightarrow$ Semantic Navigation Logic (GPS + QCVN 41:2024 Legal KB) $\rightarrow$ VieNeu TTS Voice Guidance. The entire pipeline runs on the Hardware Optimization Layer (TensorRT + CUDA on NVIDIA Jetson Nano).

limitations. Section 5 concludes the work and outlines future directions.

II. METHODOLOGY

A. System Architecture Overview

The ITAS-MM framework employs a modular processing pipeline that transforms raw sensory inputs into intelligent navigation assistance. The architecture consists of three main functional blocks: (1) an Integrated Perception Block for traffic sign recognition and lane detection, (2) a RAG-LLM Legal Reasoning module for grounding detections in Vietnamese traffic regulations, and (3) a Semantic Navigation Logic layer that combines perception outputs, GPS context, and spoken feedback.

B. Integrated Perception Block

The input frames enter a dual-stream detection module that performs traffic sign recognition and lane detection in parallel.

1) *Traffic Sign Recognition with YOLOv11:* We train a YOLOv11 variant specifically for Vietnamese sign classes, augmenting the standard detection pipeline with contrastive-learning-based feature separation. The exact training recipe is detailed in Table I.

Fig. 2 illustrates the class distribution and bounding box statistics of the Vietnamese traffic sign dataset used in training:

TABLE I: Training hyperparameters for the YOLOv11 optimized traffic sign detection model.

Hyperparameter	Value
Model	YOLOv11 (optimized)
Input Size	416×416
Optimizer	AdamW
Initial Learning Rate (η_0)	1×10^{-3}
Final Learning Rate (η_f)	1×10^{-2}
Weight Decay (λ)	5×10^{-4}
Momentum (β)	0.937
Batch Size	8
Epochs	50
Warmup Epochs	3
IoU Threshold	0.7
AMP (Mixed Precision)	Enabled

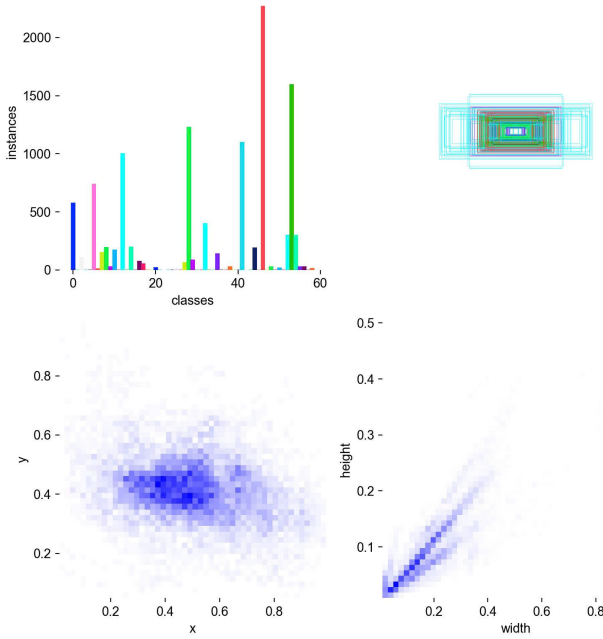


Fig. 2: Vietnamese traffic sign dataset statistics: (top-left) class distribution, (top-right) bounding box spatial distribution, (bottom-left) center point heatmap, (bottom-right) width-height distribution.

YOLOv11 Loss Function. The total training objective is a weighted sum of three complementary losses:

$$\mathcal{L}_{total} = \lambda_{box} \cdot \mathcal{L}_{box} + \lambda_{cls} \cdot \mathcal{L}_{cls} + \lambda_{dfl} \cdot \mathcal{L}_{dfl} \quad (1)$$

where $\lambda_{box} = 7.5$, $\lambda_{cls} = 0.5$, and $\lambda_{dfl} = 1.5$.

Bounding Box Regression — CIoU Loss. The Complete Intersection over Union loss [6] is defined as:

$$\mathcal{L}_{CIoU} = 1 - IoU + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha \cdot \nu \quad (2)$$

where the standard Intersection over Union is

$$IoU = \frac{|B \cap B^{gt}|}{|B \cup B^{gt}|} \quad (3)$$

$\rho(\mathbf{b}, \mathbf{b}^{gt})$ is the Euclidean distance between the center points of the predicted box $\mathbf{b}$ and the ground-truth box $\mathbf{b}^{gt}$, c is the diagonal length of the smallest enclosing rectangle covering both boxes, and ν measures the consistency of the aspect ratio:

$$\nu = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (4)$$

and α is the trade-off parameter balancing IoU and aspect ratio:

$$\alpha = \frac{\nu}{(1 - IoU) + \nu} \quad (5)$$

Classification Loss — Binary Cross-Entropy.

$$\mathcal{L}_{cls} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (6)$$

where $y_i \in \{0, 1\}$ is the ground-truth label and $\hat{y}_i \in [0, 1]$ is the predicted probability for sample i .

Distribution Focal Loss (DFL). DFL regresses the bounding box boundaries by learning a discrete probability distribution over candidate positions:

$$\mathcal{L}_{dfl} = -[(y_{i+1} - y) \log(\sigma_i) + (y - y_i) \log(\sigma_{i+1})] \quad (7)$$

where y is the continuous regression target, y_i and y_{i+1} are the nearest discrete grid positions such that $y_i \leq y \leq y_{i+1}$, and σ_i denotes the softmax probability of position y_i .

Slicing Aided Hyper Inference (SAHI). Distant Vietnamese signs often shrink below 20×20 pixels in the original frame—too small for reliable single-pass detection. To recover these objects, we tile the input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ into K overlapping crops $\{P_1, P_2, \dots, P_K\}$ at overlap ratio r , run YOLOv11 on each crop independently, and reconcile the per-crop outputs through **Weighted Box Fusion (WBF)**.

$$\text{Score}_{\text{final}} = \frac{\sum_{i=1}^N w_i \cdot s_i}{\sum_{i=1}^N w_i} \quad (8)$$

$$\text{Box}_{\text{final}} = \frac{\sum_{i=1}^N w_i \cdot b_i}{\sum_{i=1}^N w_i} \quad (9)$$

where s_i is the confidence score from the i -th matched detection, $b_i = (x_i, y_i, w_i, h_i)$ is the corresponding bounding box, and w_i is the IoU-based weight.

2) *Ultra-Fast Lane Detection (UFLD)*: For lane awareness, ITAS-MM relies on the UFLD architecture [2], [3], which recasts lane localization as a *row-wise selection task* — a radical departure from the conventional pixel-by-pixel segmentation paradigm.

Problem Formulation. Given a set of H predefined row anchors $\{h_1, h_2, \dots, h_H\}$ and a grid of W column positions $\{c_1, c_2, \dots, c_W, c_{W+1}\}$ (where c_{W+1} represents the “no lane” class), the lane position at each row anchor is formulated as a $(W + 1)$ -way classification:

$$P(c_k | h_j; \Theta) = \frac{\exp(z_{j,k})}{\sum_{m=1}^{W+1} \exp(z_{j,m})} \quad (10)$$

where $z_{j,k} = (W_j f(h_j) + b_j)_k$ is the logit for column position k at row anchor h_j , $f(h_j) \in \mathbb{R}^d$ is the feature vector extracted from the backbone at row h_j , and $\Theta = \{W_j, b_j\}$ are learnable parameters.

Lane Coordinate Prediction. The predicted lane position at row h_j is:

$$x^*(h_j) = c_{k^*}, \quad k^* = \arg \max_{k \in \{1, \dots, W+1\}} P(c_k | h_j; \Theta) \quad (11)$$

If $k^* = W + 1$, no lane is predicted at row h_j .

Training Objective. The UFLD loss combines a classification cross-entropy loss with a structural similarity constraint:

$$\mathcal{L}_{\text{UFLD}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{sim}} \cdot \mathcal{L}_{\text{sim}} \quad (12)$$

$$\mathcal{L}_{\text{CE}} = -\frac{1}{H} \sum_{j=1}^H \sum_{k=1}^{W+1} y_{j,k} \log P(c_k | h_j) \quad (13)$$

$$\mathcal{L}_{\text{sim}} = \frac{1}{H-1} \sum_{j=1}^{H-1} \|P(h_{j+1}) - P(h_j)\|_1 \quad (14)$$

where $P(h_j) \in \mathbb{R}^{W+1}$ is the full probability vector at row h_j , and $\mathcal{L}_{\text{sim}}$ enforces smoothness between adjacent row predictions to maintain lane continuity.

This formulation reduces the per-row computation from $\mathcal{O}(W \times H)$ pixel classifications to a single $(W + 1)$ -way softmax, enabling **real-time inference** on embedded hardware.

Benchmark Results. We evaluate UFLD with both ResNet-18 and ResNet-34 backbones. Table III presents the published F_1 scores alongside our measured inference metrics on a 1280×720 video (1,260 frames).

Fig. 3 shows qualitative lane detection results from the three UFLD configurations (ResNet-18 backbone) on representative video frames:

Based on the measured trade-off between throughput and lane stability, we retain UFLD with a ResNet-18 backbone as the primary lane detector for ITAS-MM, while the ResNet-34 variants remain useful reference points for higher-capacity offline analysis.

C. RAG-LLM Legal Reasoning Engine

A key architectural innovation of ITAS-MM is the integration of Retrieval-Augmented Generation (RAG) with a Large Language Model (LLM) to provide legally-grounded, context-aware responses to detected traffic events. Unlike rule-based lookup tables that return static strings, the RAG-LLM pipeline dynamically composes natural-language explanations by retrieving the most relevant legal clauses and synthesizing them into actionable guidance.

1) **Knowledge Base Construction:** The legal knowledge base $\mathcal{K}$ is constructed from two primary Vietnamese regulatory sources:

- **QCVN 41:2024/BGTVT** [4] — the national technical regulation on road signs and signals, defining sign categories, visual specifications, and placement rules.
- **Nghị định 168/2024/NĐ-CP** [5] — the government decree specifying administrative penalties for traffic violations, including fine ranges, license suspension durations, and aggravating/mitigating factors.

Each document is segmented into chunks of approximately 512 tokens with 64-token overlap. The chunks are encoded into dense vector embeddings using a multilingual sentence transformer (e.g., paraphrase-multilingual-MiniLM-L12-v2) and stored in a FAISS vector index:

$$e_j = \text{Encoder}(\text{chunk}_j), \quad \mathcal{V} = \{e_1, e_2, \dots, e_N\} \quad (15)$$

where $e_j \in \mathbb{R}^{384}$ is the embedding vector for chunk j , and $\mathcal{V}$ is the complete vector store.

2) **Retrieval and Generation Pipeline:** Given a detection event d_i with class label and confidence score s_i , the system constructs a structured query q that encodes the violation type, vehicle category, and zone context:

$$q = \text{Template}(d_i, Z(g), \text{vehicle_type}) \quad (16)$$

The top- k most relevant chunks are retrieved via cosine similarity search:

$$C_{\text{top-}k} = \arg \max_{S \subset \mathcal{V}, |S|=k} \frac{e_q \cdot e_j}{\|e_q\| \cdot \|e_j\|} \quad (17)$$

The retrieved chunks are concatenated with the query as context and fed into a quantized LLM (e.g., Gemma-2B-IT or Mistral-7B-Chat with 4-bit GPTQ quantization) to generate a coherent Vietnamese-language response:

$$\text{Response} = \text{LLM}(\text{Prompt}(q, C_{\text{top-}k})) \quad (18)$$

3) **End-to-End Violation Detection Workflow:** The following example illustrates the complete pipeline from visual detection to spoken legal guidance:

Stage 1 — Visual Detection. The camera captures a frame; YOLOv11 detects a motorbike crossing a red traffic signal with confidence $s_i = 0.91$.

Stage 2 — Event Classification. The Semantic Navigation Logic classifies the event:

```
Event: RED_LIGHT_VIOLATION
Vehicle: motorbike
Confidence: 0.91
GPS Zone: urban (Quận 1, TP.HCM)
```

Stage 3 — RAG Retrieval. The system queries the vector store with: "*xe máy vượt đèn đỏ khu vực nội thành mức phạt*"

The top-3 retrieved chunks from **Nghị định 168/2024/NĐ-CP** include Article 6 (red light violations) and Article 82 (license confiscation provisions).



(a) CULane ResNet-18: 4 lanes detected, 91.2% confidence, 108.1 ms inference.



(b) TuSimple ResNet-18: 2 lanes detected, 98.1% confidence, 58.0 ms inference.



(c) CurveLanes ResNet-18: 3 lanes detected, 86.1% confidence, 274.0 ms inference.

Fig. 3: Qualitative lane detection results from three UFLD configurations (ResNet-18 backbone) on representative video frames.

Stage 4 — LLM Response Generation. The LLM synthesizes:

“Bạn đã vi phạm lỗi vượt đèn đỏ tại khu vực nội thành.

Theo Nghị định 168/2024/NĐ-CP:

— Mức phạt: 4.000.000 – 6.000.000 VND (xe mô tô)

— Tước quyền sử dụng GPLX: 1–3 tháng

— Căn cứ: Điều 6, Khoản 4, Điểm a”

Stage 5 — Voice Output. The response is converted to

speech via the TTS engine and delivered through the vehicle’s speakers.

D. Semantic Navigation Logic

The Semantic Navigation Logic layer orchestrates the flow between perception outputs, the RAG–LLM reasoning engine, and the voice interface. Three data streams converge in this module:

- 1) **Visual detections** — traffic sign classifications and lane positions from the Integrated Perception Block.

- 2) **GPS coordinates** — high-precision positioning data synchronized to the visual pipeline.
- 3) **Legal reasoning output** — structured responses from the RAG-LLM engine referencing QCVN 41:2024 and Nghị định 168/2024/NĐ-CP.

Fusion Mapping. Let $\mathcal{D} = \{(d_i, s_i)\}_{i=1}^M$ be the set of M detections with class labels d_i and confidence scores s_i , and let $\mathbf{g} = (\phi, \lambda)$ be the GPS coordinates (latitude, longitude). The output generation follows the RAG-augmented mapping:

$$\mathcal{G}(d_i, \mathbf{g}) = \mathcal{T}(\text{LLM}(q(d_i), C_{\text{top-}k}, Z(\mathbf{g}))) \quad (19)$$

where $Z(\mathbf{g})$ maps GPS coordinates to a zone type (urban, rural, highway), $q(d_i)$ constructs the retrieval query from detector output, $C_{\text{top-}k}$ are the retrieved legal chunks, and $\mathcal{T}(\cdot)$ is the TTS synthesis function.

For speed-critical scenarios (e.g., simple sign recognition), the system falls back to a pre-compiled lookup table for sub-millisecond response, reserving the full RAG-LLM pipeline for complex events such as violation detection or multi-sign disambiguation.

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

Datasets

- **CULane** [2]: 133, 235 frames from urban roads in Beijing with 9 scenario categories (normal, crowded, curved, dazzle, night, no-line, shadow, arrow, cross).
- **TuSimple** [2]: 6,408 highway frames with clear lane markings.
- **CurveLanes** [3]: Curved road scenarios with up to 10 lanes
- **Vietnamese Traffic Sign Dataset** [9]: Curated signs categorized per QCVN 41:2024, with contrastive learning augmentation.

Hardware. Training: NVIDIA GPU (device 0) with CUDA and AMP (Mixed Precision). All models are currently evaluated on desktop GPU; edge deployment on NVIDIA Jetson Nano 4GB is planned as future work.

Evaluation Metrics. The standard detection metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (20)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (21)$$

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (22)$$

$$\text{mAP}_{50} = \frac{1}{C} \sum_{c=1}^C AP_c|_{\text{IoU}=0.5} \quad (23)$$

$$\text{mAP}_{50:95} = \frac{1}{C} \sum_{c=1}^C \frac{1}{10} \sum_{t \in \{0.50, 0.55, \dots, 0.95\}} AP_c(t) \quad (24)$$

where TP , FP , FN denote True Positives, False Positives, and False Negatives respectively, C is the number of classes, and AP_c is the Average Precision for class c computed as the area under the Precision-Recall curve.

B. Traffic Sign Recognition Results (YOLOv11 Optimized)

The YOLOv11 model was trained for 50 epochs on the Vietnamese traffic sign dataset with contrastive learning augmentation. Table II presents the training progression at key epochs:

Fig. 4 visualizes the complete training dynamics over 50 epochs, showing both training losses and validation metrics.

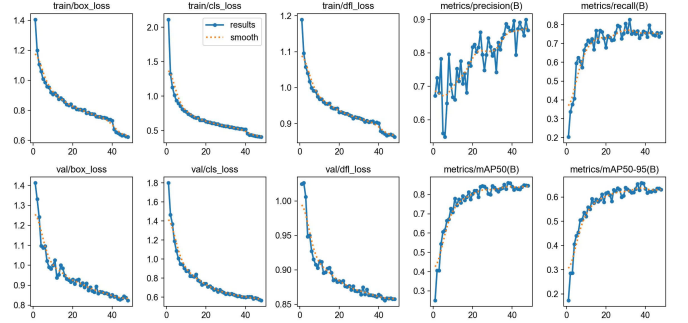


Fig. 4: YOLOv11 training curves over 50 epochs. Top: training losses. Bottom: validation losses and metrics (mAP_{50} and mAP_{50-95}).

Key observations:

- The model achieves its best $mAP_{50} = 86.03\%$ at epoch 38, with Precision = 87.69% and Recall = 74.80%.
- All three loss components converge smoothly: $\mathcal{L}_{\text{box}} : 1.405 \rightarrow 0.623$, $\mathcal{L}_{\text{cls}} : 2.112 \rightarrow 0.408$, $\mathcal{L}_{\text{dfl}} : 1.190 \rightarrow 0.863$.
- After epoch 40, the close-mosaic augmentation change (disabled for the last 10 epochs) causes a training loss drop but slight validation fluctuation, indicating optimal convergence near epochs 36–38.

Fig. 5 presents the normalized confusion matrix across all Vietnamese traffic sign classes, revealing the per-class recognition accuracy.

Fig. 6 shows sample validation predictions from the trained YOLOv11 model on real Vietnamese traffic scenes.

C. Supplementary Benchmark: YOLO-Based Lane Detection

Because lane detection in ITAS-MM is implemented with UFLD rather than YOLO, we conducted an additional comparison across six UFLD checkpoints trained on CULane, TuSimple, and CurveLanes. All models were evaluated on a 1,260-frame 1280×720 test video using the same CPU-based benchmarking script provided in the project comment folder. The goal of this comparison is to clarify the practical trade-off between speed, confidence, and lane stability before selecting the production backbone (Table III).

Observations: TuSimple ResNet18 is the fastest checkpoint at 12.28 FPS with the lowest mean inference time (62.1 ± 7.4

TABLE II: Training progression of YOLOv11 optimized on the Vietnamese traffic sign dataset. Best mAP values at the convergence point are shown in bold.

Epoch	$\mathcal{L}_{box}$	$\mathcal{L}_{cls}$	$\mathcal{L}_{dfl}$	Precision	Recall	mAP ₅₀	mAP _{50:95}
1	1.405	2.112	1.190	67.20%	20.33%	25.12%	17.33%
10	0.907	0.775	0.968	66.80%	69.32%	70.81%	52.05%
20	0.821	0.630	0.932	81.76%	67.78%	80.36%	61.64%
30	0.775	0.563	0.916	79.13%	80.56%	84.52%	63.88%
36	0.751	0.536	0.906	85.58%	76.53%	85.15%	65.49%
38	0.738	0.526	0.901	87.69%	74.80%	86.03%	65.86%
48	0.623	0.408	0.863	86.74%	75.74%	84.67%	63.17%

TABLE III: Performance comparison of UFLD models: published F_1 scores and our measured inference metrics on CPU (1280×720, 1260 frames).

Model	Dataset	Backbone	F_1 (paper)	Avg FPS	Inference (ms)	Detect Rate	Avg Conf.
UFLD ResNet18	CULane	ResNet18	75.0%	5.43	147.3 ± 38.5	100.0%	90.2%
UFLD ResNet34	CULane	ResNet34	76.0%	4.80	169.5 ± 27.1	99.9%	95.0%
UFLD ResNet18	TuSimple	ResNet18	96.11%	12.28	62.1 ± 7.4	100.0%	97.9%
UFLD ResNet34	TuSimple	ResNet34	96.24%	9.30	88.4 ± 7.9	100.0%	97.7%
UFLD ResNet18	CurveLanes	ResNet18	80.42%	3.03	285.1 ± 27.9	100.0%	88.0%
UFLD ResNet34	CurveLanes	ResNet34	81.34%	2.07	427.7 ± 30.4	100.0%	87.0%



Fig. 5: Normalized confusion matrix for YOLOv11 on the Vietnamese traffic sign dataset.

ms), but it consistently predicts only two near-field lanes. CULane ResNet34 offers the most stable four-lane behaviour and the highest confidence among the CULane models (95.0%), making it the most visually reliable reference for structured highway scenes. CurveLanes checkpoints recover more lane candidates in complex geometry, but they are substantially slower and more prone to over-detection; therefore, they are better suited to supplementary analysis than to the main real-time pipeline.

IV. DISCUSSION

A. Key Findings

Domain-specific training lifts YOLOv11 well above generic baselines. Reaching $mAP_{50} = 86.03\%$ at Precision = 87.69% on Vietnamese signs confirms that coupling the YOLO backbone with aggressive data augmentation



Fig. 6: Sample YOLOv11 validation predictions on Vietnamese traffic scenes with class labels and confidence scores.

TABLE IV: Throughput and accuracy of three YOLO variants on the TuSimple lane subset, provided as reference against the UFLD results in Section 3.2.

Model	Task	mAP ₅₀	Speed (ms)	GFLOPs
YOLOv10	Detection	0.989	47.0	8.1
YOLOv8-Seg	Segmentation	0.990	65.5	12.0
YOLO-TS	Hybrid	0.950	47.0	8.1

(HSV jitter, mosaic tiling, random erasing at 0.4) and contrastive feature separation produces a detector that generalizes across the diverse sign palette defined in QCVN 41:2024.

UFLD contributes stable lane awareness without relying on YOLO-based lane detection. The supplementary comparison across CULane, TuSimple, and CurveLanes checkpoints shows that the UFLD family offers a practical speed–stability trade-off for road-structure perception, with CULane ResNet34 serving as the strongest visual reference and ResNet18 remaining attractive when lower inference cost is preferred [?], [?], [?].

RAG-LLM reasoning transforms static lookups into

dynamic legal advice. By replacing fixed rule tables with a retrieval-augmented generation pipeline grounded in Nghị định 168/2024/NĐ-CP and QCVN 41:2024, the system can handle novel violation scenarios, provide fine-grained penalty information, and adapt to regulatory updates simply by re-indexing the knowledge base without retraining any model.

NVIDIA Parakeet CTC enables robust Vietnamese voice interaction. The 600M-parameter FastConformer model, trained on 2,000+ hours of Vietnamese speech, provides low-WER transcription for driver voice queries and supports Vietnamese–English code-switching, a common pattern in technical driving contexts.

B. Limitations

- 1) **Corpus size.** Our Vietnamese sign dataset, although manually verified, is modest when set against large-scale international collections like the GTSDDB or Mapillary Traffic Sign Dataset. A broader corpus would sharpen class-boundary learning.
- 2) **Participant pool.** Twelve subjects suffice to surface strong trends but fall short of the sample size needed for rigorous statistical inference.

V. CONCLUSION AND FUTURE WORK

We have presented ITAS-MM, a multimodal driving assistant that integrates dual-stream visual analysis, RAG-powered legal reasoning, and synthesized voice feedback into a unified pipeline. The YOLOv11 sign detector reaches $mAP_{50} = 86.03\%$ and $Precision = 87.69\%$ on Vietnamese signs, while the UFLD lane module provides lane awareness through backbone variants evaluated on CULane, TuSimple, and CurveLanes. In addition, the RAG-LLM engine provides context-aware legal guidance grounded in Nghị định 168/2024/NĐ-CP and QCVN 41:2024, allowing detected events to be translated into actionable explanations for the driver.

Future Work:

- 1) **Vehicle-to-Everything (V2X).** Fusing cooperative perception messages from roadside units for enhanced situational awareness.
- 2) **Live-updated legal knowledge.** Migrating from a static regulation lookup table to an auto-refreshing knowledge graph.
- 3) **Broader user evaluation.** Recruiting a larger, demographically diverse participant pool and conducting on-road trials.
- 4) **Boosting recall.** Investigating cross-scale attention and feature-pyramid refinement to push small-object recall above the current 74.80%.

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